

Multifunctional Pd@UiO-66 Catalysts for Continuous Catalytic Upgrading of Ethanol to *n*-Butanol

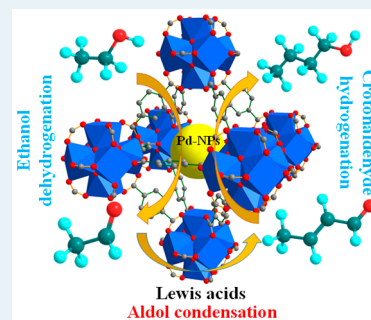
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Supporting Information

ABSTRACT: UiO-66-encapsulated nanopalladium (Pd@UiO-66) catalysts were used as highly efficient and stable catalysts for continuous catalytic upgrading of ethanol to *n*-butanol. The best Pd@UiO-66 catalyst exhibited 49.8% of ethanol conversion, 48.6% of selectivity toward *n*-butanol, and thereby 24.2% of *n*-butanol yield at relatively low temperature (523 K) and pressure (2 MPa) during a 200 h long-term evaluation. The high catalytic activity and selectivity of Pd@UiO-66 catalyst are primarily ascribed to the close synergy of highly dispersed Pd nanoparticles and coordinatively unsaturated Zr sites on Zr₆ nodes of UiO-66, as active centers for dehydrogenation/hydrogenation and aldol condensation, respectively; however, the high stability of the catalyst is mainly attributed to the electrostatic attraction of Pd nanoparticles with Zr₆ nodes and the confinement effect of the cavities of UiO-66.

KEYWORDS: multifunctional catalysts, Pd nanoparticles, metal–organic framework, Lewis acids, ethanol upgrading



Catalytic upgrading of bioethanol to *n*-butanol has currently been receiving great attention both in scientific and industrial fields, in view of the advantages of *n*-butanol over bioethanol as a sustainable fuel.^{1,2} It is generally accepted that *n*-butanol can be produced from ethanol by the Guerbet pathway (Scheme S1, Supporting Information), which consists of four tandem steps: ethanol dehydrogenation, aldol condensation of acetaldehyde, dehydration of the aldol, and hydrogenation of crotonaldehyde.^{3–6} Solid materials with Lewis acid and base sites, such as metal oxides or mixed metal oxides,⁷ alkali metal-modified zeolites⁸ and hydroxyapatite (HAP)^{9–11} were widely employed as active catalysts for this reaction. However, these traditional catalyst systems normally showed poor *n*-butanol yields (<10%) and worked at high temperatures and pressures (>573 K, > 6 MPa). Incorporation of metal nanoparticles into these acid–base materials was proven to significantly improve *n*-butanol yield, mainly because of two reasons: (1) increasing ethanol dehydrogenation and crotonaldehyde hydrogenation activity of catalysts, which were accomplished primarily through the Meerwein–Ponndorf–Verley (MPV) reaction over the previous catalysts;¹² and (2) promoting the aldol condensation of acetaldehyde by improving acid–base properties of catalysts. For instance, M–CeO₂/AC (M = Cu, Fe, Co, Ni and Pd) catalysts developed by our group exhibited a selectivity dependence on the synergy of metal sites and CeO₂ basic sites.^{13,14} The highest selectivity to *n*-butanol (67.6%) was achieved over Pd–CeO₂/AC catalyst at 523 K and 2.0 MPa, in which Pd metals improved the basicity of CeO₂ because of their strong capabilities of hydrogen activation and spillover. However, long-term experiments showed that metal-supported

catalysts suffered from metal sintering and then the decay of activities.^{13–15}

On the other hand, Lewis base sites or acid–base pairs are generally considered as effective active sites for the aldol condensation of acetaldehyde, which is the rate-determining step in catalytic upgrading of ethanol to *n*-butanol.¹⁶ It was previously assumed that strong Lewis base sites were required because of their high activity toward the removal of α -H of acetaldehyde,³ as in the case of M–CeO₂/AC catalysts. However, later studies revealed that these sites were quickly deactivated due to acetaldehyde polycondensation, and the further reaction actually proceeded over Lewis acid–weak base pairs.¹⁷ Although the strength of Lewis base sites was weakened, the catalytic activity of aldol condensation over Lewis acid–weak base pairs was enhanced.⁷ The reason for this observation was proposed that Lewis acids could stabilize ethoxide intermediates^{18,19} and activate carbonyl group of acetaldehyde,^{6,17} which points to the importance of Lewis acids in the reaction. However, no further investigation on the role of Lewis acids has been reported, which is probably due to the fact that it is difficult to measure the contribution of Lewis acids without the interference of Lewis bases over aforementioned traditional catalysts. Thus, it is imperative to have approaches to understand the behavior of Lewis acids during the course of reaction, which is expected to further facilitate the aldol condensation (the rate-determining step) and consequently the catalytic upgrading of ethanol to *n*-butanol.

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Table 1. Catalytic Performance of Pd@UiO-66 Catalysts for Continuous Catalytic Upgrading of Ethanol to *n*-Butanol^a

catalysts	conversion (%)	selectivity (%)						butanol yield (%)
		acetaldehyde	ether	butyraldehyde	butanol	>C ₄ products ^b	others ^c	
UiO-66	2.6	3.4	92.9	1.1	0.3	3.3	-	trace
0.5 wt %Pd@UiO-66	36.1	5.1	2.5	4.1	60.2	21.0	7.1	21.7
1 wt %Pd@UiO-66	46.9	3.8	4.0	3.9	50.6	31.9	5.9	23.8
2 wt %Pd@UiO-66	49.9	3.4	6.5	3.5	50.1	27.9	8.6	25.0
3 wt %Pd@UiO-66	43.8	4.4	8.0	4.2	45.5	28.1	9.8	20.0
2 wt %Pd/UiO-66	32.4	7	8.7	5.2	39.4	28.7	11	12.8
2 wt %Pd@UiO-66 ^d	49.8	3.5	5.8	4	48.6	28.9	9.2	24.2

^aConversion, selectivity, and yield were obtained at steady-state; reaction conditions: 0.5 g of catalyst, 523 K, 2 MPa, LHSV = 4 mL/(h·g·cat), N₂/ethanol(v/v) = 250:1. ^b>C₄ products mainly include 2-ethylbutyraldehyde, hexaldehyde, 2-ethylbutanol, and 1-hexanol; ^cOther products include ethyl acetate, 1,1-diethoxyethane, butyl acetate, etc.; ^d200 h of time on stream.

Metal–organic frameworks (MOFs), in particular, Zr-based MOFs with outstanding stability,^{20–22} have been extensively investigated for Lewis acid catalyzed reactions such as citronellal cyclization,²³ esterification,²⁴ transfer hydrogenation,²⁵ and cross-aldol condensation.²⁶ Besides as Lewis acids in catalysis, the combination of MOFs with metal nanoparticles (Metal-NPs@MOFs) could exhibit more advantages as heterogeneous catalysts.^{27–30} For instance, the uniform and small cavities of MOFs restrict the growth of metal NPs and then afford more active metal sites. Additionally, the existence of ordered multiple active sites (i.e., metal NPs, Lewis acid sites on nodes, and functional groups of organic ligands) in one MOFs enables the flexible design of multifunctional catalysts for complicated reactions, such as tandem or domino (cascade) reactions.^{31,32} More importantly, in this Metal-NPs@MOFs, the synergetic effect of the components is significant, which enables active sites to cooperate in a concerted way for chemical transformation, thus achieving high activity and selectivity.^{33,34}

In the present study, UiO-66-encapsulated nanopalladium (Pd@UiO-66) catalysts were prepared by a simple impregnation-reaction method (IRM)^{35–38} and applied in a tandem reaction, the catalytic upgrading of ethanol to *n*-butanol. These catalysts exhibited excellent performance under milder reaction conditions because of the close synergy of confined Pd nanoparticles (Pd-NPs) and coordinatively unsaturated Zr sites (Zr-CUSs) on nodes of UiO-66, which act as active centers for dehydrogenation/hydrogenation and aldol condensation, respectively. To the best of our knowledge, it is the first time to present that (1) MOFs-supported metal catalysts have been applied in the continuous conversion of ethanol to *n*-butanol; (2) Lewis acids (Zr-CUSs) could have better performance than Lewis bases or Lewis acid–base pairs for aldol condensation; (3) the highest *n*-butanol yield (24.2%) and superior stability (200 h) that have been reported over heterogeneous catalysts under milder reaction conditions (523 K and 2.0 MPa); and (4) MOFs could be used in reactions with the temperature as high as 523 K and the pressure up to 2.0 MPa.

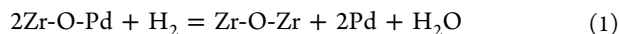
The catalytic performance of various Pd@UiO-66 catalysts is displayed in Table 1. UiO-66 support itself showed only 2.6% ethanol conversion, 0.3% selectivity to *n*-butanol but 92.9% selectivity to diethyl ether. The incorporation of Pd-NPs dramatically improved the catalytic performance of MOFs-based catalysts. Pd@UiO-66 catalysts with various Pd loadings all exhibited higher ethanol conversions (36.1–49.9%), selectivities to *n*-butanol (45.5–60.2%), and *n*-butanol yields (20.0–25.0%) than previously reported heterogeneous cata-

lysts. The best catalyst, 2 wt %Pd@UiO-66, exhibited 49.9% of ethanol conversion, 50.1% of selectivity to *n*-butanol, and 25.0% of *n*-butanol yield in a 12 h evaluation. In contrast, the 2 wt %Pd/UiO-66 catalyst prepared by a wet impregnation method not only had lower ethanol conversion (32.4%) but also lower selectivity to *n*-butanol (39.4%) than Pd@UiO-66 catalysts.

In order to shed light on the mechanism of the catalytic performance of Pd@UiO-66 catalysts, XRD and TEM were first employed to characterize the catalysts before and after 12 h of reactions. As shown in XRD analysis (Figure S2, Supporting Information) and Table S1 (Supporting Information), the crystallinity of UiO-66 and other studied catalysts were all well-retained, and no characteristic diffraction peaks of metal Pd were detected for the Pd@UiO-66 catalysts before and after 12 h of reactions, implying the high dispersion of metal Pd. These results were further confirmed by TEM characterizations (Figures S3–S5, Supporting Information). It can be found that Pd-NPs with a mean diameter of 1.8 nm (Figure S3A, Supporting Information) and 2.2 nm (Figure S4A, Supporting Information) were evenly dispersed over the reduced 0.5 and 2 wt %Pd@UiO-66 catalyst, respectively. The encapsulation of Pd-NPs inside the crystalline grains of UiO-66 was further verified by tilting the TEM grids and imaging the same Pd-NPs from different angles. Even for the catalyst with high Pd loading (3 wt %Pd@UiO-66), the average particle size of Pd-NPs was only 2.7 nm, though a few Pd-NPs with larger size appeared on the external surface of UiO-66 (Figure S5A, Supporting Information). After 12 h of reaction, Pd-NPs remained highly dispersive with the size less than 3.1 nm over each Pd@UiO-66 catalyst (Figures S3–S5B and Table S1, Supporting Information). In contrast, for the reduced 2 wt %Pd/UiO-66 catalyst, larger and unevenly distributed Pd-NPs with the size of 5.0 nm were observed dominantly on the external surface of UiO-66; moreover, after 12 h of reaction, these Pd-NPs grew up and aggregated together (Figure S6, Supporting Information).

UiO-66 consists of 12-connected Zr₆(μ₃-O)₄(μ₃-OH)₄ inorganic nodes coordinated by 1,4-benzene-dicarboxylate (BDC) linkers.^{20–22} The Zr₆ nodes bearing μ₃-OH groups endow UiO-66 with great potential to anchor guest active species in an analogous manner to traditional oxide supports (e.g., silica or alumina).^{35–38} In the present study, Pd@UiO-66 catalysts were prepared by means of IRM, in which Pd ions were anchored on both inner and outer surfaces of UiO-66 support via the reaction of Pd(acac)₂ with μ₃-OH groups on the Zr₆ nodes in the form of Zr–O–Pd bonds. The formation of these bonds can be confirmed by FT-IR results that the

intensity of absorption band at 3674 cm^{-1} assigned to stretching vibrations of $\mu_3\text{-OH}$ groups on the Zr_6 nodes was significantly reduced after the IRM (Figure S7, Supporting Information). During the catalyst reduction, the Zr-O-Pd bonds could break with the production of oxygen vacancies and Zr-CUSs on the Zr_6 nodes via the following reaction:



The increase in the concentration of Zr-CUSs can be corroborated by the decrease in the atomic molar ratio of O/Zr from XPS analysis (Table S2, Supporting Information). Moreover, the reduced Pd-NPs (nucleophilic) were prone to interact with nearby formed Zr-CUSs (electrophilic) through electrostatic attraction, thereby stabilizing Pd-NPs . The higher BE value of $\text{Pd } 3d$ over the $2\text{ wt \%Pd@UiO-66}$ catalyst than that over the $2\text{ wt \%Pd/UiO-66}$ catalyst (335.9 vs 335.6 eV , Table S2, Supporting Information) implies a stronger electrostatic attraction, which can account for better stability of Pd-NPs over the $2\text{ wt \%Pd@UiO-66}$ catalyst, as demonstrated by XRD and TEM results. This implies the IRM is an efficient method for incorporating metals onto the Zr_6 nodes, and the incorporated Pd-NPs were stable both in the process of catalyst reduction and subsequent reaction. Besides, the good thermal stability of the Pd-NPs can be also attributed to the uniform and small cavities of UiO-66 limiting the migration and aggregation of Pd-NPs .^{39,40} In short, the high dispersion of Pd-NPs over the Pd@UiO-66 catalysts stems from the interaction of Pd precursors with $\mu_3\text{-OH}$ groups of the Zr_6 nodes, by which Pd-NPs were successfully introduced inside crystalline grains of UiO-66 . Further electrostatic attraction of Pd-NPs with Zr_6 nodes and the confinement effect of the cavities of UiO-66 improved the thermal stability of Pd-NPs .

To clarify the nature of active sites for aldol condensation of acetaldehyde, the acid–base properties of UiO-66 support, Pd/UiO-66 , and Pd@UiO-66 catalysts were examined using NH_3 - (Figure 1A) and CO_2 -TPD (Figure 1B). Two types of

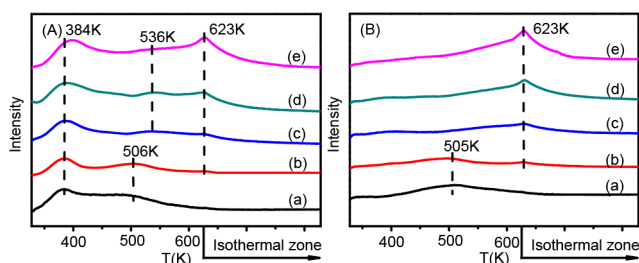


Figure 1. (A) NH_3 -TPD and (B) CO_2 -TPD profiles: (a) UiO-66 ; (b) $2\text{ wt \%Pd/UiO-66}$; (c) $0.5\text{ wt \%Pd@UiO-66}$; (d) $2\text{ wt \%Pd@UiO-66}$; (e) $3\text{ wt \%Pd@UiO-66}$.

Lewis acid sites on the Zr_6 nodes of UiO-66 were involved in the NH_3 desorption, namely, 7- and 6-fold-coordinated Zr sites formed after the removal of hydroxyl groups or partial framework linkers upon thermal treatment.^{41,42} For the UiO-66 support (Figure 1A(a)), the peak at 384 K is attributed to the NH_3 desorption from the former acid sites, while the weak and broad peak centered at 506 K belongs to the latter ones.^{43,44} The $2\text{ wt \%Pd/UiO-66}$ catalyst displayed a similar NH_3 desorption pattern as that on UiO-66 , implying no significant difference between them in terms of the quantity and strength of Lewis acid sites. It is reasonable considering that there was no chemical interaction between Pd precursors

and UiO-66 support (e.g., forming Zr-O-Pd bonds) in the catalyst preparation. Over Pd@UiO-66 catalysts as shown in Figure 1A(c–e), the low temperature peak (384 K) slightly shifted to higher temperatures with the increase in Pd loading, and a desorption peak at high temperature (536 K) relative to that on UiO-66 (506 K) was observed. These higher temperatures required for NH_3 desorption are actually ascribed to the exposure of more Zr-CUSs with lower coordination number (6-fold Zr-CUSs). Another desorption peaks at 623 K observed for the various Pd -containing catalysts (Figure 1A(b–e)) should result from a slight decomposition of UiO-66 support in the presence of metal Pd as demonstrated by TG-MS results of $2\text{ wt \%Pd@UiO-66}$ catalyst (Figures S8–9, Supporting Information). With the increase of Pd loading, the peak at 623 K intensified, indicating a higher decomposition degree of the UiO-66 support. These observations point out that metal Pd would benefit the generation of 6-fold Zr-CUSs (via the formation and subsequent cleavage of Zr-O-Pd bonds), but also cause the loss of framework linkers and consequently the instability of catalysts at high Pd loading. Thus, the $2\text{ wt \%Pd@UiO-66}$ catalyst appears to be optimal among all Pd@UiO-66 catalysts, when balancing the activity and stability of catalysts.

As shown in Figure 1B, there was a CO_2 desorption peak at 505 K for the UiO-66 support, which is ascribed to CO_2 desorption from Lewis base sites, probably $\mu_3\text{-O}$ on the Zr_6 nodes. This peak was not observable over Pd@UiO-66 catalysts, indicating the absence of Lewis base sites after the introduction of Pd-NPs (Figure 1B(c–e)). As discussed in NH_3 -TPD results, the addition of metal Pd increased the amount of 6-fold Zr-CUSs on the Zr_6 nodes. Following this, the decrease in basicity of $\mu_3\text{-O}$ is thus ascribed to the short distance between $\mu_3\text{-O}$ and 6-fold Zr-CUSs , which resulted that more electronic density was concentrated in the Zr-O bond rather on the oxygen.⁴² In contrast, the intensity of CO_2 desorption peak over the $2\text{ wt \%Pd/UiO-66}$ catalyst only slightly changed relative to that over UiO-66 , implying most of Lewis base sites still retained. This phenomenon is also reasonable considering the number of Zr-CUSs over the $2\text{ wt \%Pd/UiO-66}$ catalyst was almost the same as that over UiO-66 .

According to the results of NH_3 - and CO_2 -TPD, it is conclusive that Lewis acid sites were dominant over Pd@UiO-66 catalysts, which was further confirmed by FT-IR experiments of pyridine and CO_2 adsorption (Figure S10, Supporting Information). As mentioned previously, Lewis base sites have been considered to play crucial roles in the aldol condensation of acetaldehyde; however, Pd@UiO-66 catalysts exhibited much better performance (Table 1). This implies that Lewis acids could act as more active centers than Lewis bases or Lewis acid–base pairs for the reaction, which was also verified by FT-IR results of acetaldehyde adsorption and reaction over the $2\text{ wt \%Pd@UiO-66}$ catalyst whose Lewis acid and base sites were separately poisoned with pyridine and CO_2 (Figure S11, Supporting Information).

Kinetic analysis (Figures S12–S18 and Tables S3–S7, Supporting Information) over the $2\text{ wt \%Pd@UiO-66}$ catalyst reveals the apparent activation energy for ethanol dehydrogenation, aldol condensation of acetaldehyde and crotonaldehyde hydrogenation was 37.7 , 55.8 , and $4.2\text{ kJ}\cdot\text{mol}^{-1}$, respectively, indicating that the aldol condensation is still the decisive step in these tandem steps. However, it is notable that the apparent activation energy of aldol condensation was much lower than

our reported state-of-the-art Cu-CeO₂/AC catalysts (70.1 kJ·mol⁻¹).¹³ Kinetic analysis further corroborates that Lewis acid sites (Zr-CUSs) were very active toward the aldol condensation of acetaldehyde. Hence, the high *n*-butanol yield (25.0%) is ascribed to the cooperation of Pd-NPs and Zr-CUSs on the Zr₆ nodes of UiO-66 support. The reaction mechanism for this synergy in the conversion of ethanol to *n*-butanol is depicted in Scheme S2 (Supporting Information).

Regarding the industrial applications of Pd@UiO-66 catalysts, they are required to have not only excellent catalytic activity and selectivity to target products but also a good long-term stability. Fortunately, the 2 wt %Pd@UiO-66 catalyst exhibited superior stability in a 200 h on-stream evaluation, showing a promising prospect of industrial applications (Figure 2A). Ethanol conversion only slightly changed from 51.4% to

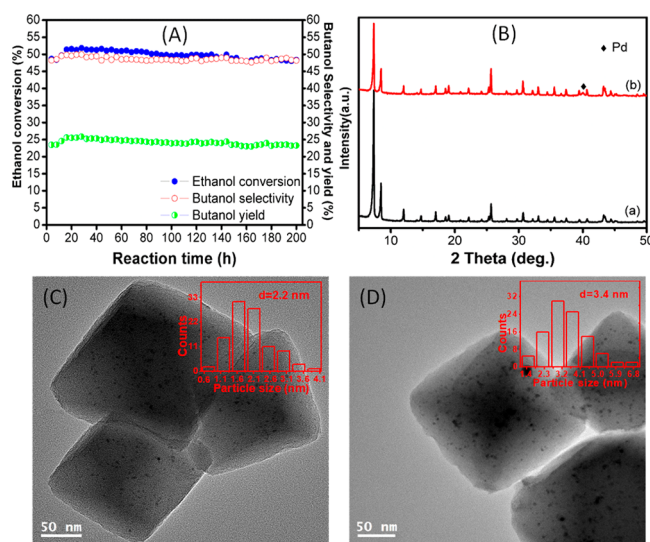


Figure 2. Long-term stability (A) of the 2 wt % Pd@UiO-66 catalyst, and its XRD patterns and TEM images (B-a), (C) before, and (B-b), (D) after 200 h of reaction.

49.8% for the initial and final ethanol conversion, respectively, while the selectivity to *n*-butanol kept constant (48.6%) during the whole process. XRD and TEM analyses for the 2 wt %Pd@UiO-66 catalyst before and after 200 h of reaction were also shown in Figure 2. From Figure 2B, it can be seen that the crystal structure of the 2 wt %Pd@UiO-66 catalyst after 200 h reaction was almost the same as that of the one before reaction, except for the appearance of a very diffuse peak at 40.1° of metal Pd. The integrity of the 2 wt %Pd@UiO-66 catalyst can be also confirmed from TEM images in Figure 2C,D, although there was a slight growth of Pd-NPs from 2.2 to 3.4 nm, which might account for the minor change of ethanol conversion. Further research on stabilizing the Pd-NPs in long-term operations is being carried out in our lab.

In summary, UiO-66-encapsulated nanopalladium (Pd@UiO-66) catalysts could be successfully prepared by a simple impregnation-reaction method and applied as highly efficient catalysts for the catalytic upgrading of ethanol to *n*-butanol. It was found that Pd nanoparticles were favorable for ethanol dehydrogenation and crotonaldehyde hydrogenation, while coordinatively unsaturated Zr sites acting as Lewis acids were very active toward the aldol condensation of acetaldehyde. In this concerted way, the 2 wt %Pd@UiO-66 catalyst exhibited 49.9% of ethanol conversion, 50.1% of selectivity toward *n*-

butanol, and 25.0% of *n*-butanol yield at 523 K and 2 MPa in the 12 h reaction. This catalyst also displayed good stability in the 200 h long-term evaluation, which could be ascribed to the electrostatic attraction of Pd nanoparticles with Zr₆ nodes and the confinement effect of the cavities of UiO-66. The present study expands the reaction scope of MOFs-based materials in the field of heterogeneous catalysis, especially in complicated tandem reactions or reactions with high reaction temperatures or pressures.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acscatal.8b04014.

Experimental methods, characterization data, and kinetic analysis (Schemes S1–S2, Figures S1–S18, and Tables S1–S7) (PDF)

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Notes

The authors declare no competing financial interest.

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